Lesson 5.5 Guided Notes

Session 1 · Volume of Rectangular Prisms

I can find the volume of a rectangular prism by multiplying length, width, and height. · 6.G.A.2

Word / Spanish	Meaning — complete the blank	Example
volume / volumen	The amount of _____ inside a solid.	V = l × w × h
unit cube / cubo unitario	A cube with every edge _____ unit long.	1 cm³
base area / área de la base	The area of the _____ face of a prism.	B = l × w
cubic units / unidades cúbicas	Units used to measure _____.	cm³, in³, ft³

Key idea

One layer of cubes holds length × _____ cubes.

The height tells how many _____ stack up.

V = l × w × h, or V = _____ × h.

Worked example

A box is 6 cm long, 3 cm wide, and 4 cm tall. Find its volume.

One layer: 6 × 3 = _____ cubes.

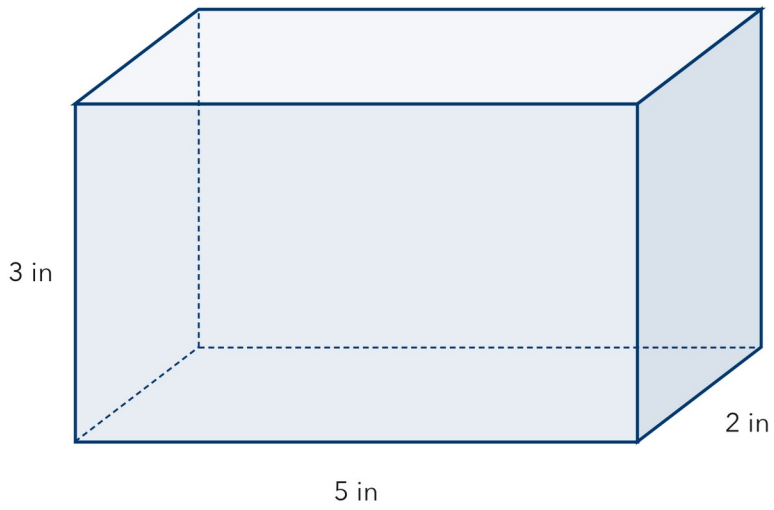
The height is 4, so there are _____ layers.

Multiply: 18 × 4 = _____.

Answer: V = _____ cm³

You try

1. Find the volume of the prism.



_____

2. Find each volume.
a. 10 m × 2 m × 3 m _____ m^3
b. B = 25 ft^2, h = 4 ft _____ ft^3

_____

Sum it up: Volume uses cubic units because ___.

_____

Session 2 · Missing Fractional Edges

I can find a missing edge of a prism and find volumes with fractional edge lengths. · 6.G.A.2

Word / Spanish	Meaning — complete the blank	Example
edge / arista	A _____ where two faces of a solid meet.	length, width, height
base area / área de la base	The _____ of the length and width of the base.	B = l × w
inverse operations / operaciones inversas	Operations that _____ each other.	÷ undoes ×

Key idea

V = l × w × h works with _____ too.

To find a missing edge, _____ the volume by the product of the other two edges.

Check: the three edges should _____ to the volume.

Worked example

A box holds 72 cm³. It is 6 cm long and 4 cm wide. Find its height.

Known edges: 6 × 4 = _____ cm².

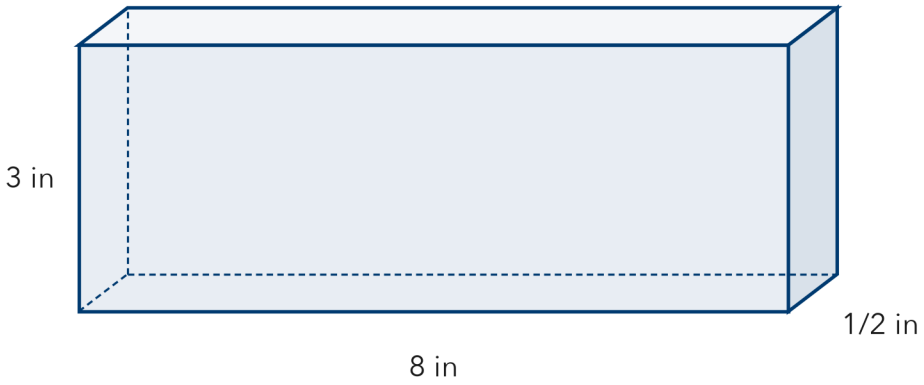
Divide: 72 ÷ 24 = _____.

Check: 6 × 4 × 3 = _____.

Answer: height = _____ cm

You try

1. Find the volume of the prism.



_____

2. Find each value.
a. V = 60 m³, base 3 m by 4 m; h = _____ m
b. V of 1 1/2 ft × 2 ft × 3 ft _____ ft³

_____

Sum it up: To find a missing edge, I ___ because ___.

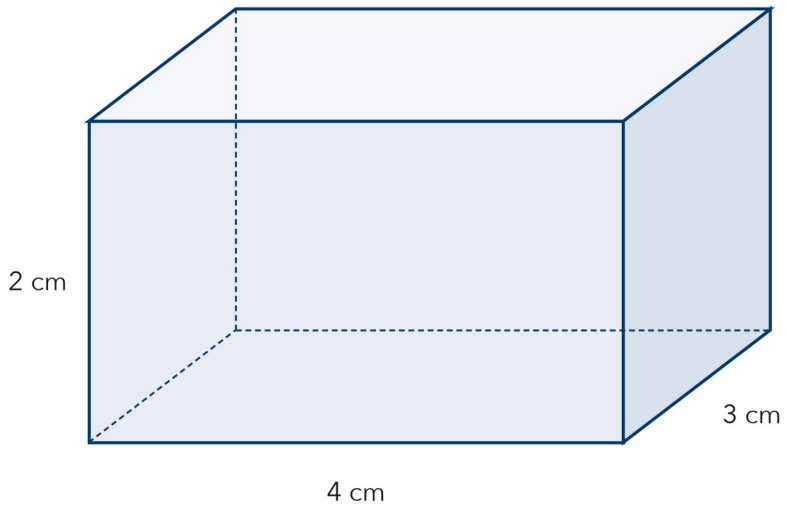
_____

Session 1 · Model, reason, and explain

Volume of Rectangular Prisms · 6.G.A.2

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A prism is 4 cm by 3 cm by 2 cm. Draw one layer of unit cubes, then show how layers make the volume. Explain square versus cubic units.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: calls base area the volume or uses square units for volume.. Explain what changes and what stays true.

_____

_____

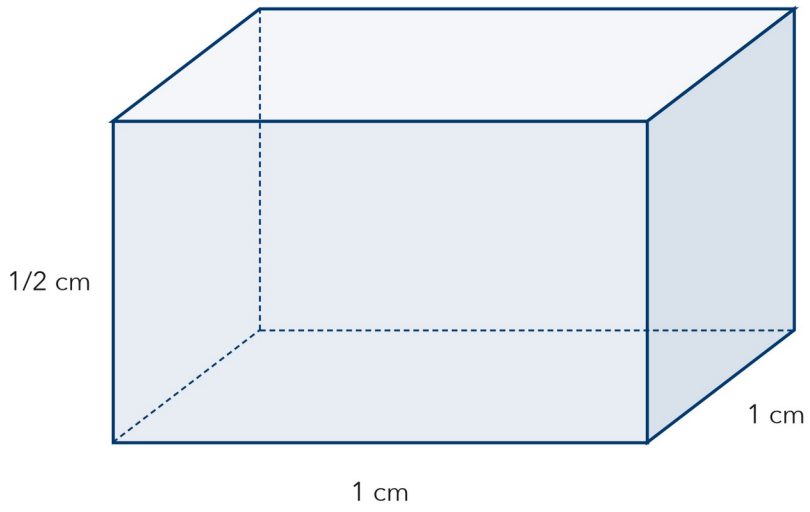
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Missing Fractional Edges · 6.G.A.2

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A prism is 1 cm by 1 cm by 1/2 cm. Pack it with cubes whose edges are 1/2 cm. How many cubes fit, and what volume does each represent?



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: counts small cubes as full cubic centimeters.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?